

Papers discussed in the the last meeting:

1. Threshold displacement energies in refractory high-entropy alloys

Abstract -

Refractory high-entropy alloys show promising resistance to irradiation, yet little is known about the fundamental nature of radiation-induced defect formation. Here, we simulate threshold displacement energies in equiatomic MoNbTaVW using an accurate machine-learned interatomic potential, covering the full angular space of crystal directions. The effects of local chemical ordering are assessed by comparing results in random and short-range-ordered MoNbTaVW. The average threshold displacement energy in the random alloy is 44.3 ± 0.15 eV, and slightly higher, 48.6 ± 0.15 eV, in the short-range-ordered alloy. Both are significantly lower than in any of the constituent pure metals. We identify the mechanisms of defect creation and find they are mainly dependent on the masses of the recoiling and colliding elements. Low thresholds are generally found when heavy atoms (W, Ta) displace and replace the lightest atoms (V). The average threshold energies, when separated by recoiling element, are consequently ordered inversely according to their mass, opposite to the trend in the pure metals where W has by far the highest thresholds. However, the trend in the alloy is reversed when considering the cross sections for defect formation in electron irradiation, due to the mass-dependent recoil energies from the electrons.